Formal Verification of Quantum Codes with Transversal Diagonal Gates in Lean 4

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arXiv:2510.20728 · github.com/LionSR/lean-qec

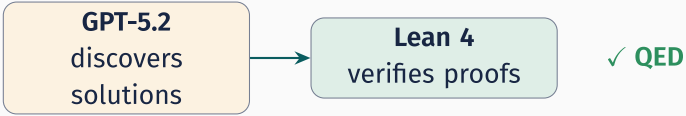
Outline

This Talk

Starting point: arXiv:2510.20728 catalogued 14,116 distance-2 codes with transversal diagonal gates via the SSLP pipeline.

Three new contributions:

1. **Lean 4 formalization** of the full SSLP pipeline — machine-checked proofs, zero `sorry`
2. **Analytical families**: one proof covers *infinitely many* codes
3. **Distance-3 generalization**: exact solutions for $((7, 2, 3))$ codes discovered by GPT-5.2, formally verified in Lean



Transversal Gates & the SSLP Pipeline

Transversal Diagonal Gates

Physical gate with angle vector $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{Z}_m^n$:

$$U(\mathbf{a}, m) = \bigotimes_{i=1}^{n} \begin{pmatrix} 1 & 0 \\ 0 & \omega_m^{a_i} \end{pmatrix}, \qquad \omega_m = e^{2\pi i/m}$$

Action on computational basis: $U(\mathbf{a}, m)|x\rangle = \omega_m^{\langle \mathbf{a}, x\rangle}|x\rangle$ where $\langle \mathbf{a}, x\rangle = \sum_i a_i x_i$.

Define **residue classes** $C_b(\mathbf{a}) = \{x \in \{0,1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$.

Eigenstate condition

$\mathrm{supp}(|j_L\rangle) \subseteq C_{s_j}(\mathbf{a}) \implies U(\mathbf{a}, m)|j_L\rangle = \omega_m^{s_j}|j_L\rangle$

The SSLP Pipeline

Subset-**S**um **L**inear **P**rogramming: systematic code construction.

Step 1: SS	→	**Step 2: LP**	→	**Step 3: Full KL**
Support partition into residue classes		Z-type KL feasibility (linear program)		X/Y-type constraints (nonlinear)
combinatorial		*convex*		*non-convex*

Step 2 is an LP $\Rightarrow$ fast rejection of infeasible parameters.

For distance 2: our residue-shift screen makes Step 3 *automatic*.

For distance 3: we solve Step 3 via exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$.

Step 1: Subset-Sum Support Partition

Residue classes $C_b(\mathbf{a}) = \{x \in \{0,1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$ partition $\{0,1\}^n$.

- **Disjoint**: $C_b \cap C_{b'} = \emptyset$ for $b \neq b' \Rightarrow$ automatic orthogonality
- **Logical action**: $U\,|j_L\rangle = \omega_m^{S_j}\,|j_L\rangle$
- **Projective order**: $\mathcal{O} = m \,/\, \gcd(m, S_1 - S_0, \dots)$

Example: $n{=}5$, $m{=}7$

$\mathbf{a}{=}(1,1,2,2,2)$, $(S_0, S_1) = (0,4)$

$C_0 = \{00000, 01111, 10111\}$

$C_4 = \{00011, 00101, \dots\}$ (6 elts)

$\mathcal{O} = 7/\gcd(7,4) = 7$

Step 2: Z-Type KL as Linear Program

Probabilities $p_{j,x} = |c_{j,x}|^2$ on support of $|j_L\rangle$.

Z-type KL: $\langle j_L|Z_i|j_L\rangle = t_i$ independent of j becomes

$$\boxed{\sum_{x \in C_{S_j}} (-1)^{x_i} p_{j,x} = t_i \qquad \forall\, i \in [n], \forall\, j}$$

Linear in $p_{j,x}$ with constraints:

- $\sum_{x \in C_{S_j}} p_{j,x} = 1$
- $p_{j,x} \geq 0$

Standard LP feasibility.

Geometric view: sign vectors $v(x) = ((-1)^{x_1}, \ldots, (-1)^{x_n})$ are vertices.
Feasibility $=$ nonempty $\bigcap_j \mathrm{conv}(V_j)$.

Residue-Shift Screen: Distance 2 for Free

Flipping bit i shifts the residue by $\pm a_i$:

$$\langle \mathbf{a}, x \oplus e_i \rangle \equiv \langle \mathbf{a}, x \rangle \pm a_i \pmod{m}$$

Residue-Shift Screen

$S_j - S_k \not\equiv \pm a_i \pmod{m}$ for all $i \in [n]$, all $j \neq k \implies$ union support has $d_H \geq 2$.

When the screen passes, all X/Y-type KL vanish automatically.

SS partition + residue-shift screen $\xrightarrow{\text{finite check}}$ Z-type KL (LP feasibility) $\longrightarrow$ ✓ **Distance 2**

Binary Dihedral Symmetry (K=2)

Binary dihedral group BD_{2m}:

- Generated by $Z(2\pi/m)$ and $\hat{X} = -iX$
- $|\mathrm{BD}_{2m}| = 4m$; $\quad m{=}8 \Rightarrow$ transversal T
- Transversal $\overline{X} = X^{\otimes n}$: $|1_L\rangle = X^{\otimes n}\,|0_L\rangle$
- Support: $S_1 = \bar{S}_0$ (bitwise complements)
- Amplitudes: $c_{1,\bar{x}} = c_{0,x}$

BD angle condition

$\sum_{j=1}^{n} a_j \equiv -1 \pmod{m}$

***Z*-balance**

Complement symmetry forces $\sum_{x\in S_0}(-1)^{x_i} p_x = 0$ for all i.

Only $|0_L\rangle$ needs specification.

Distance-2 Results

Catalogue: 14,116 Distance-2 Codes

[arXiv:2510.20728]

Systematic search via SSLP:

- $K \in \{2, 3, 4\}$, $n \leq 6$, $m \leq 18$
- Residue-shift screen for $d{=}2$
- LP solver + rational reconstruction

14,116

certified $((n, K, 2))$ codes

Highlights

- Orders $\mathcal{O} = 2$ through 18 at $n = 6$
- Mostly **nonadditive**
- Includes $K = 3, 4$ codes

Analytical families

Family I: $C_0 = \{0^n, 1^n\}$,
closed-form $p(0^n) = 1 - s/m$
✓verified in Lean for all (n, m, s)

Analytical Families: One Proof, Infinitely Many Codes

Find a **closed-form pattern**, prove it *once*, instantiate everywhere.

Family I: the simplest codes

Support $C_0 = \{0^n, 1^n\}$ — just two basis states.

$$|0_L\rangle = \sqrt{p}\ |0\cdots0\rangle + \sqrt{1-p}\ |1\cdots1\rangle$$

Z-balance forces p **uniquely**: $p = 1 - s/m$.

Valid for *all* $(n, m, \mathbf{a})$ passing the screen.

λ-families

Larger supports, free parameter λ:

- λ-Family II: $|C_0| = 4$
- λ-Family V: $|C_0| = 6$
- 5 families, each a **universally quantified** Lean theorem

One Lean proof $\Leftrightarrow$ infinitely many verified codes

Worked Example: $((5,2,2))$ Code

$n{=}5$, $\mathbf{a}{=}(1,1,2,2,2)$, $m{=}7$, $(S_0,S_1)=(0,4)$

SS: $C_0=\{00000, 01111, 10111\}$, C_4: 6 elements

Screen: $4\not\equiv \pm1, \pm2 \pmod 7$ ✓

LP: $p_{00000}=3/7$, $p_{01111}=p_{10111}=2/7$

$\langle Z_i\rangle = 3/7$ $(i{=}1,2)$, $-1/7$ $(i{=}3,4,5)$

Lean: one-line proof

```
def myCode : ExampleData where
  n := 5; m := 7; K := 2
  a := ![1, 1, 2, 2, 2]
  S := ![0, 4]
  supp := ![supp0, supp1]
  prob := ![prob0, prob1]

theorem verified : myCode.OK :=
  by native_decide
```

SS + **screen** + **LP** $\Rightarrow$ certified $((5,2,2))$ code, order $\mathcal{O}=7$. Build: <1 s.

Lean 4 Formalization

What is Lean 4?

A **proof assistant** based on dependent type theory:

- Every proof is a program; the **kernel** type-checks it
- Compiles $\Rightarrow$ proof is correct. No trust required.
- **Mathlib**: 180K+ formalized theorems

Key tactic: `native_decide`

- Proposition is `Decidable`? Compile to native code $\to$ run $\to$ `true`
- Kernel accepts: proof complete
- "All 49,152 KL constraints hold?" $\to$ `true` $\to$ QED

Landmark projects

- Freiman–Ruzsa conj. (Tao)
- Liquid Tensor Experiment
- Sphere eversion
- This work: quantum codes

Why Lean for QEC?

14K codes: too many for humans.

AI-found solutions: trust but verify.

Formal proof $=$ **certainty**.

Lean 4 Codebase: `lean-qec`

Core Library (ss/)

`BitString.lean`

`Basic.lean`

`DiagonalAction.lean`

`ResidueShiftScreen.lean`

`ZTypeKL.lean`

`BDSymmetry.lean`

`Verify.lean`

`FamilyI/` (6 files)

`LambdaV2/` (5 families)

Distance-3 Modules

`Pauli.lean`

`FullKL.lean`

`QSqrt23i.lean`

`FullKLEval.lean`

`Distance3Verify.lean`

`BD16Defs.lean`

Codebase

$\sim$4,300 lines, $\sim$30 modules

0 `sorry`, 0 warnings

Examples

`Examples/D2/`

- `Ex522.lean` ($(5, 2, 2)$)
- `Ex622.lean` ($(6, 2, 2)$)

`Examples/D3/`

- `BD16v1.lean` through
- `BD16v6a.lean` (8 codes)

Build

3,091 jobs

0 errors · 0 warnings

Formalization: Key Theorems

`Dact_eq_global_of_SS`

$\mathrm{supp}(|j_L\rangle) \subseteq C_{S_j} \Rightarrow U\,|j_L\rangle = \omega_m^{S_j}\,|j_L\rangle$

`no_hamming1_neighbor_of_screen`

Screen condition $\Rightarrow d_H(x, y) \geq 2$ across classes

$\Rightarrow$ all X/Y-type KL vanish

`ExampleData.OK`

Bundles four checks:

1. SS support condition
2. Residue-shift screen
3. Probability normalization
4. Z-type KL matching

All `Decidable` $\Rightarrow$ `native_decide` in < 1 s

Technique: `BitString n` $=$ `Fin` $n \to$ `Bool` has 2^n elements. All quantifiers over finite support are decidable $\Rightarrow$ Lean compiles to native code, executes, returns `true` $\Rightarrow$ kernel accepts as proof.

Distance-3 Extension

The Distance-3 Barrier

Why was distance 2 "easy"?

- Residue-shift screen kills all X/Y-type KL automatically
- Only Z-type survives $\Rightarrow$ linear in probabilities $p_x \geq 0$
- No complex amplitudes needed

Why is distance 3 harder?

- Must check all weight-≤ 2 Paulis: $Z_iZ_j, X_iX_j, X_iY_j, \ldots$
- X, Y **flip bits** $\Rightarrow$ couples different basis states
- KL becomes **quadratic** in $c_x \in \mathbb{C}$
- Relative phases matter

The gap: distance-2 $=$ convex optimization over $\mathbb{R}_{\geq 0}$.
Distance-3 $=$ nonlinear system over $\mathbb{C}$. At $n{=}7$: $3 \times 2^{14} = 49{,}152$ constraints.

Core challenge: solve $\sim$50,000 nonlinear constraints over $\mathbb{C}$, then verify solutions with mathematical certainty.

Distance 3: Three Key Ideas

1. Most constraints vanish.
$S_0 \cap (S_0 \oplus z_P) = \emptyset$
$\Rightarrow$ sum is empty $\Rightarrow$ 0.
BD16v1: 25 / 49,152
$>$ 99.5% are free.

2. Exact number field.
Amplitudes in
$\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$
$=$ 8 rational coefficients.
`DecidableEq`, no floats.

3. AI + Lean.
GPT-5.2 solves the
$\sim$25–150 surviving
constraints symbolically.
Lean verifies *exactly*.

Intersection test filters 99.5% → GPT-5.2 solves remaining system → Lean verifies every identity

Why Most KL Constraints Vanish

$\langle 0_L|P|0_L\rangle$ sums over pairs where x and $x \oplus z_P$ are *both* in S_0.

Disjoint $\Rightarrow$ empty sum $\Rightarrow$ automatically 0

$$S_0 \cap (S_0 \oplus z_P) = \emptyset \implies \langle 0_L|P|0_L\rangle = 0.$$

Why so effective

S_0 is **sparse:** $|S_0| = 7$ out of $2^7 = 128$. Random shifts almost never land back inside S_0. Only 25 / 49,152 survive (BD16v1).

In Lean: `klIsAutomatic` is `Decidable` — checked by `native_decide`.

From Constraints to Solutions: GPT-5.2

After intersection test, ∼25–150 surviving constraints form a **quadratic system** in $c_x \in \mathbb{C}$.

The problem:

- Unknowns: c_x for each $x \in S_0$
- $\sum_x \bar{c}_x \, c_{x'} \cdot (\text{phase}) = 0$
- Non-convex $\Rightarrow$ hard for standard solvers

GPT-5.2's approach:

- Given S_0, probabilities, constraint eqns
- Exploits symmetries, groups terms
- Returns **closed-form** amplitudes in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$

Constraint equations → **GPT-5.2** symbolic solve → **Lean 4** `native_decide` ✓ **QED**

Exact Arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$

$m{=}8 \Rightarrow \omega_8 = \frac{1+i}{\sqrt{2}}$. Amplitudes involve $\sqrt{2}$, $\sqrt{3}$ from normalization.

Representation (`QSqrt23i`)

Each element $= (q_0, \dots, q_7) \in \mathbb{Q}^8$:

$$(q_0 + q_1\sqrt{2} + q_2\sqrt{3} + q_3\sqrt{6}) + (\cdots)\, i$$

Multiplication closed.
Equality *decidable* (rational coefficients).

BD16v1 amplitudes (GPT-5.2)

$c_0 = \frac{1}{4}$ $\quad c_5 = \frac{1}{4}$
$c_1 = c_2 = \frac{\sqrt{3}}{4}$ $\quad c_6 = \frac{i}{2}$
$c_3 = c_4 = \frac{\sqrt{2}}{4}$

In Lean: $c_1 = (0, 0, \frac{1}{4}, 0,\ 0, 0, 0, 0)$ ✓

No floating point. Every operation exact over $\mathbb{Q}$. `DecidableEq` $\Rightarrow$ `native_decide` verifies each KL identity.

Two-Layer Verification Architecture

Layer 1 (combinatorial, on probabilities)

SS + BD angle + normalization + Z-balance: $\sum(-1)^{x_i} p_x = 0 \quad \forall i$

$\downarrow$ *pass*

Layer 2 (amplitude-level, on $c_x \in \mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$)

All weight-≤ 2 KL matrix elements $= 0$

(non-automatic only; rest filtered by intersection test)

Both layers are **fully decidable** — verified by `native_decide`.

No `sorry`, no axioms. Build time per code: 2–11 s.

BD16 Example: $((7,2,3))$ with a $= (1,2,2,2,2,3,3)$

m=8 (transversal T), $|S_0| = 7$.

$$4\,|0_L\rangle = |0000000\rangle + \sqrt{3}\,|0111100\rangle + \sqrt{3}\,|1001110\rangle + \sqrt{2}\,|1010101\rangle + \sqrt{2}\,|1011001\rangle + |1110010\rangle + 2i\,|010$$

```
def bd16Data : Distance3Data 7 8 where
  a := ![1, 2, 2, 2, 2, 3, 3],
  b := 0,
  supp := {s0, s1, s2, s3, s4, s5, s6},
  prob := ... -- (1,3,3,2,2,1,4)/16

theorem layer1 : bd16Data.Layer1OK
  := by native_decide
theorem layer2 : AllKLSatisfied ,supp ,amp 2
  := by native_decide
```

Check	Result
SS + BD + norm	✓ native_decide
Z-balance	✓ native_decide
Non-auto KL	25 / 49152
All 25 $= 0$	✓ native_decide

Complete $d\geq3$ proof
Build: 2.6 s

8 Verified BD16 $((7,2,3))$ Codes

All codes: $n=7$, $K=2$, $m=8$ (transversal T), verified `Layer1OK` + `AllKLSatisfied`.

File	Angle vector a	$\lvert S_0 \rvert$	λ^*	Solution type	Status
BD16v1	$(1,2,2,2,2,3,3)$	7	$\sqrt{21/8}$	2-param continuous	✓ native_decide
BD16v2	$(1,1,2,2,2,2,5)$	10	$\sqrt{33}/4$	192-elt discrete	✓ native_decide
BD16v3a	$(0,1,2,2,3,3,4)$	7	$\sqrt{51}/4$	Solution A (of 3)	✓ native_decide
BD16v3b	$(0,1,2,2,3,3,4)$	8	$\sqrt{23}/4$	Solution B (of 3)	✓ native_decide
BD16v3c	$(0,1,2,2,3,3,4)$	8	$\sqrt{30}/4$	Solution C (of 3)	✓ native_decide
BD16v4	$(1,1,1,1,3,3,5)$	13	$\sqrt{177}/8$	continuous	✓ native_decide
BD16v5	$(1,1,1,1,3,4,4)$	12	$\sqrt{62}/4$	128-elt discrete	✓ native_decide
BD16v6a	$(1,1,1,2,2,4,4)$	16	1	uniform magnitude	✓ native_decide

Remaining: 3 vectors need $\sqrt{10}$ (field extension); 2 proven impossible. 12 vectors total $\Rightarrow$ 10 feasible, 8 verified.

What the Surviving Constraints Look Like

After the intersection test, $\sim$25–150 non-automatic Pauli strings remain.

Diagonal (Z_iZ_j)

$\langle 0_L|Z_iZ_j|0_L\rangle = \sum_{x\in S_0}(-1)^{x_i+x_j}\,|c_x|^2$

Linear in probabilities $p_x = |c_x|^2$.

Already verified in **Layer 1**.

Key simplification

BD symmetry $|1_L\rangle = X^{\otimes n}\,|0_L\rangle$

$\Rightarrow$ off-diagonal eqns **pair up**:

$\langle 0|P|0\rangle = 0 \Leftrightarrow \langle 1|P|1\rangle = 0$

Only $|0_L\rangle$ amplitudes needed.

Off-diagonal ($X_iX_j, X_iY_j, \ldots$)

$$\langle 0_L|P|0_L\rangle = \sum_{\substack{x\in S_0\\ x\oplus z_P\in S_0}} \bar{c}_x\, c_{x\oplus z_P}\, \omega^{f(x,P)}$$

Quadratic in amplitudes $c_x \in \mathbb{C}$.

Phases $\omega^{f(x,P)}$ depend on x_P bits.

BD16v1 ($|S_0|=7$): 25 constraints $\to$ 7 complex unknowns with 1 normalization + 7 probability values fixed.

The BD_{16} Solution Landscape

All 12 BD_{16} ($(7,2,3)$) angle vectors (up to qubit permutation, $a_i \leftrightarrow m - a_i$):

Angle vector **a**	$\lvert S_0 \rvert$	Non-auto	Solution type	Status
$(1,2,2,2,2,3,3)$	7	25	2-param continuous	Lean ✓
$(1,1,2,2,2,2,5)$	10	94	discrete family	Lean ✓
$(0,1,2,2,3,3,4)$	7	47	3 isolated solutions	Lean ✓
$(1,1,1,1,3,3,5)$	13	145	continuous	Lean ✓
$(1,1,1,1,3,4,4)$	12	112	discrete family	Lean ✓
$(1,1,1,2,2,4,4)$	16	149	uniform magnitude	Lean ✓
$(1,1,2,2,3,3,3)$	14	134	needs $\sqrt{10}$	pending
$(1,1,1,2,3,3,4)$	12	103	needs $\sqrt{10}$	pending
$(1,1,1,2,2,3,5)$	10	86	needs $\sqrt{10}$	pending
$(1,1,2,2,2,6,1)$	5	11	proven infeasible	
$(0,1,1,2,3,3,5)$	4	4	proven infeasible	
$(0,0,1,2,3,3,6)$	4	2	numerical only	unresolved

8/10 feasible vectors verified. 3 pending need $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5},i)$ (16-dim field).

Outlook

Outlook: Batch Verification of All 14,116 Codes

The distance-2 catalogue has 14,116 $((n, K, 2))$ codes. Currently: individual Lean files for representative examples.

Scaling strategy

1. Python exports each code's data as a `.lean` file (support, probabilities, angle vector)
2. Lean's `ExampleData.OK` bundle checks SS $+$ screen $+$ normalization $+$ Z-KL
3. `native_decide` proves each in $< 1\,\mathrm{s}$
4. `lake build` compiles all 14K files

Already demonstrated

- `Ex522.lean`, `Ex622.lean`: fully verified via `native_decide`
- Build time: 0.3–0.8 s each
- At 1 s/code: $\sim$4 hours total (embarrassingly parallel)

14,116 machine-checked proofs: within reach

Outlook: Analytical Family II & Beyond

Family II: 4-element support

$C_0 = \{0^n, 1^n, e_1\bar{e}_1, \bar{e}_1 e_1\}$

Free parameter $\lambda \in [0,1] \to$ continuous family.

Lean: `familyII_OK` $\forall (n, m, s, \lambda)$ passing screens.

λ-Families A–E (verified)

5 families at $n{=}5$, each covers ∞ values.

$\Sigma_Z(\lambda) = \sum_i \langle Z_i \rangle^2 \in [\frac{1}{3}, 1]$

What universality buys

- One 100-line file $\to$ **all** instantiations
- Proves **structure**, not cases
- Classification complete for $|C_0| \leq 4$

Open: analytical families for distance 3?

Needs universal proofs over $\mathbb{Q}(\sqrt{\cdot}, i)$ with amplitude parameters — significantly harder.

Outlook: Beyond BD_{16}

BD_{16} at $n=7$ is the **first step**. The framework extends naturally:

Other BD groups

BD_{12}–BD_{36} ($m=6$–18)

Same pipeline; number field adapts to ω_m.

Zhang–Zeng: dozens of $((7, 2, 3))$ candidates

$n=8$: $T^{1/4}$

BD_{64} ($m=32$): transversal $\pi/16$.

Intersection test still filters $> 99\%$ of KL.

Exceptional groups

$2I$ ($|G|=120$): 5th level
$2T$ ($|G|=24$)
$2O$ ($|G|=48$)

Two-layer architecture still applies.

Long-term vision: a formal library of quantum codes with **machine-verified** fault-tolerance properties — AI discovers, Lean certifies.

Summary

Summary

Distance 2

- Full SSLP pipeline in Lean 4
- Residue-shift screen $\Rightarrow d \geq 2$ automatic
- 14,116 codes certified
- Analytical families: one proof $\to \infty$ codes

Codebase

$\sim$4,300 lines Lean 4, 0 `sorry`

`github.com/LionSR/lean-qec`

Distance 3 (new)

- Intersection test filters 99.5%
- GPT-5.2 $\to$ closed-form amplitudes
- Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$
- 8 BD_{16} $((7, 2, 3))$ codes: first machine-checked $d=3$ proofs

What's next

Batch 14K codes · $\mathbb{Q}(\sqrt{5})$ extension · BD_{12}–BD_{36} · $n=8$ · 2I/2T/2O

Thank you

arXiv:2510.20728 · `github.com/LionSR/lean-qec`